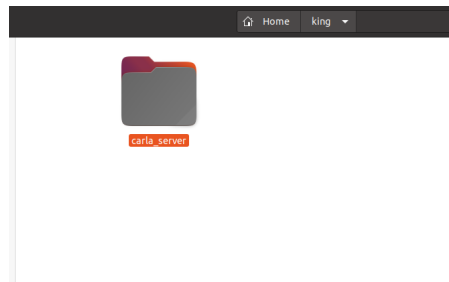
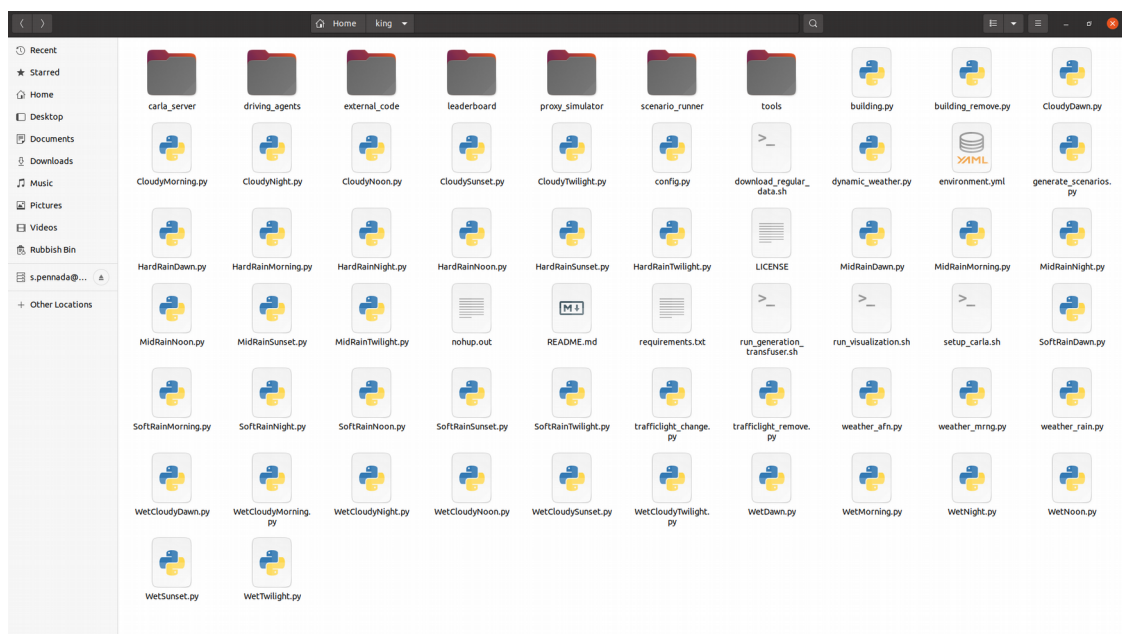


KING Replay with modifications

Download and set up CARLA 0.9.15.



Clone the repository **King-Replay** from <https://github.com/SanjeethaPennada/King-Replay>. Extract all the files into our folder (where carla is installed) as shown below. Now we have all the required files to replay king with modifications.



Now install all the required libraries listed in **requirements.txt**. Also verify the hardware and software requirements listed in **README.md** file.

Install drivers and reboot. If the appropriate version of the driver is already installed (Check with the command `nvidia-smi`), if installed should appear as below and

```
sanjeetha@dc35167:~/king$ nvidia-smi
Mon Apr 22 15:24:46 2024

+-----+
| NVIDIA-SMI 535.171.04                  Driver Version: 535.171.04      CUDA Version: 12.2     |
+-----+-----+
| GPU Name   Persistence-M| Bus-Id  Disp.A | Volatile Uncorr. ECC | |
| Fan  Temp  Perf    Pwr:Usage/Cap|         Memory-Usage | GPU-Util  Compute M. |
|           |             |          |      MIG M.     |
+-----+-----+
| 0  NVIDIA GeForce RTX 4090      Off   | 00000000:01:00.0  On   |          Default     |
| 0%    40C    P5              38W / 450W | 615MiB / 24564MiB |      0%          N/A |
+-----+-----+

Processes:
+-----+
| GPU  GI  CI | PID  Type  Process name                        | GPU Memory |
| ID   ID   ID |      |   |                           |            |
+-----+-----+
| 0  N/A  N/A | 1509  G   /usr/lib/xorg/Xorg                  | 69MiB      |
| 0  N/A  N/A | 2190  G   /usr/lib/xorg/Xorg                  | 274MiB     |
| 0  N/A  N/A | 2317  G   /usr/bin/gnome-shell                | 75MiB      |
| 0  N/A  N/A | 4795  G   /usr/lib/firefox/firefox            | 146MiB     |
+-----+-----+

sanjeetha@dc35167:~/king$
```

you can skip this step or else install them using below commands:

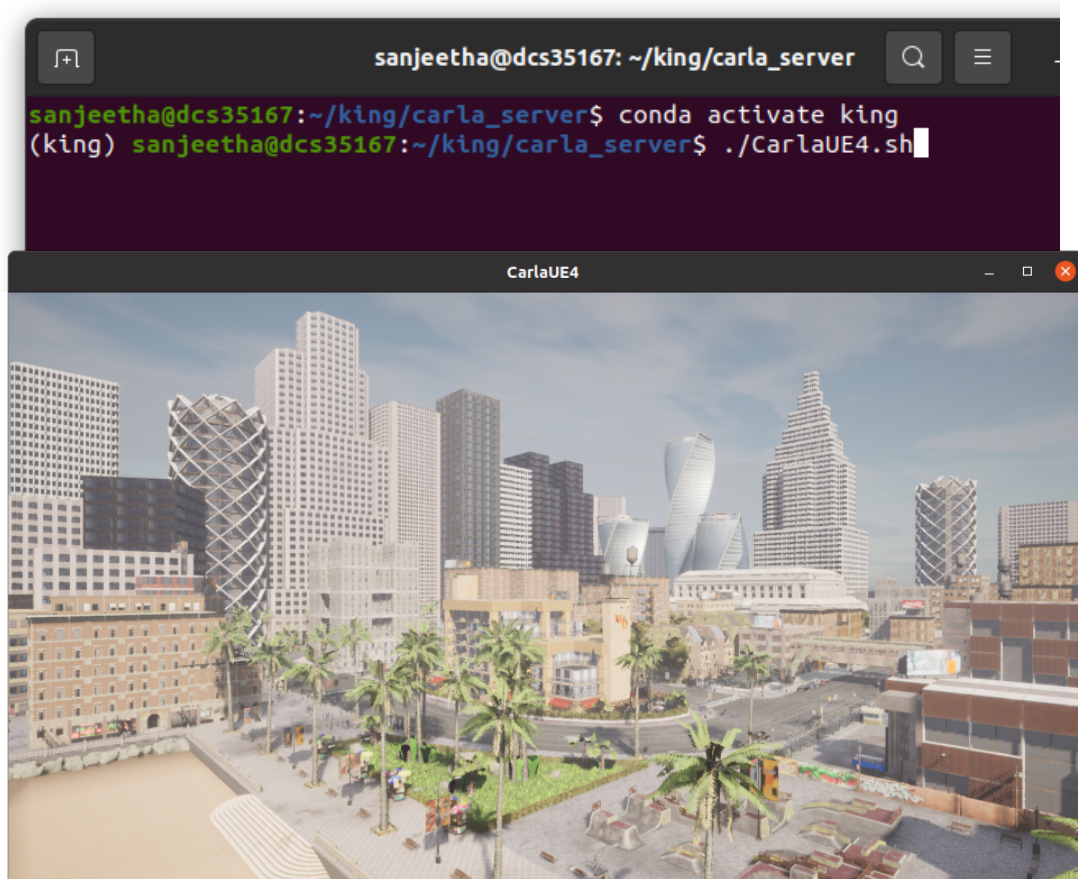
```
sudo apt update
sudo apt install ubuntu-drivers-common
sudo ubuntu-drivers autoinstall
sudo reboot
```

Install anaconda and build the environment.

```
wget https://repo.anaconda.com/archive/Anaconda3-2022.05-Linux-x86_64.sh
bash Anaconda3-2022.05-Linux-x86_64.sh
source ~/.profile
conda env create -f environment.yml
conda activate king
```

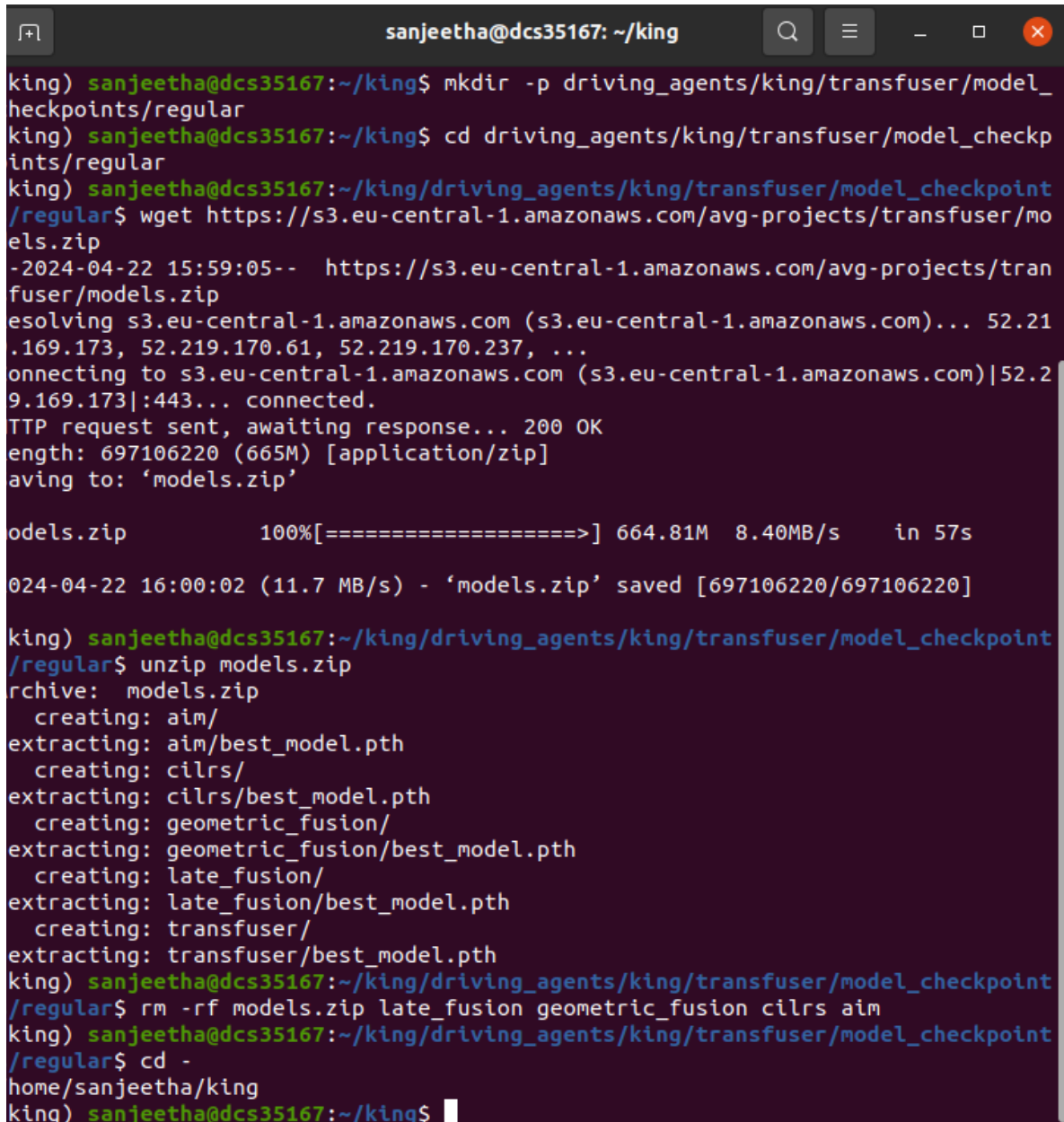
Check tworking of CARLA simulator.

Go to **carla_server** folder – right click and select open in terminal – enter **conda acticvate king** – and next enter **./CarlaUE4.sh** and CARLA is loaded successfully as shown below.



To generate scenarios for Transfuser, you need to download the model weights:

```
mkdir -p driving_agents/king/transfuser/model_checkpoints/regular
cd driving_agents/king/transfuser/model_checkpoints/regular
wget https://s3.eu-central-1.amazonaws.com/avg-projects/transfuser/models.zip
unzip models.zip
rm -rf models.zip late_fusion geometric_fusion cilrs aim
cd -
```



```
sanjeetha@dcs35167: ~/king
king) sanjeetha@dcs35167:~/king$ mkdir -p driving_agents/king/transfuser/model_
heckpoints/regular
king) sanjeetha@dcs35167:~/king$ cd driving_agents/king/transfuser/model_checkp
ints/regular
king) sanjeetha@dcs35167:~/king/driving_agents/king/transfuser/model_checkpoint
/regular$ wget https://s3.eu-central-1.amazonaws.com/avg-projects/transfuser/mo
els.zip
-2024-04-22 15:59:05-- https://s3.eu-central-1.amazonaws.com/avg-projects/tran
fuser/models.zip
resolving s3.eu-central-1.amazonaws.com (s3.eu-central-1.amazonaws.com)... 52.21
.169.173, 52.219.170.61, 52.219.170.237, ...
connecting to s3.eu-central-1.amazonaws.com (s3.eu-central-1.amazonaws.com)|52.2
9.169.173|:443... connected.
HTTP request sent, awaiting response... 200 OK
Content-Length: 697106220 (665M) [application/zip]
Saving to: 'models.zip'

models.zip          100%[=====>] 664.81M  8.40MB/s   in 57s

2024-04-22 16:00:02 (11.7 MB/s) - 'models.zip' saved [697106220/697106220]

king) sanjeetha@dcs35167:~/king/driving_agents/king/transfuser/model_checkpoint
/regular$ unzip models.zip
Archive:  models.zip
  creating: aim/
  extracting: aim/best_model.pth
  creating: cilrs/
  extracting: cilrs/best_model.pth
  creating: geometric_fusion/
  extracting: geometric_fusion/best_model.pth
  creating: late_fusion/
  extracting: late_fusion/best_model.pth
  creating: transfuser/
  extracting: transfuser/best_model.pth
king) sanjeetha@dcs35167:~/king/driving_agents/king/transfuser/model_checkpoint
/regular$ rm -rf models.zip late_fusion geometric_fusion cilrs aim
king) sanjeetha@dcs35167:~/king/driving_agents/king/transfuser/model_checkpoint
/regular$ cd -
home/sanjeetha/king
king) sanjeetha@dcs35167:~/king$
```

TransFuser generation and Scenario Replay

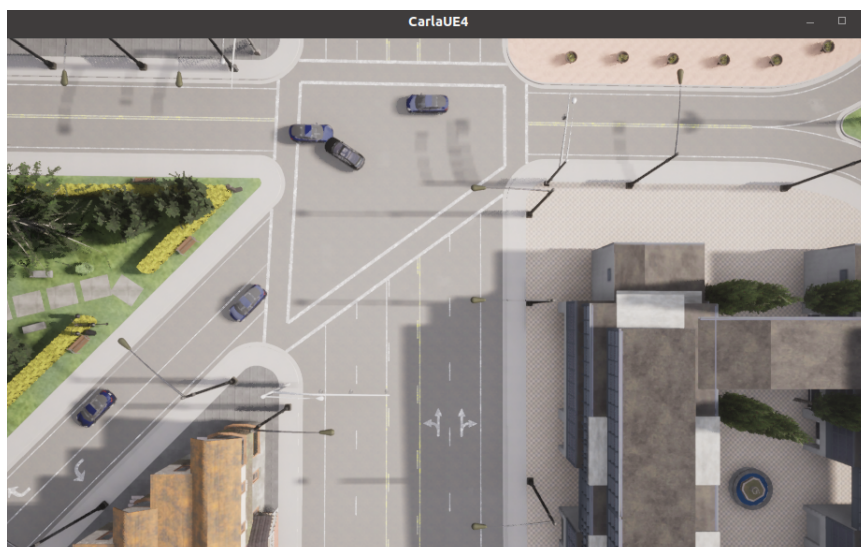
For Transfuser generation using both gradient paths, open run_generation_transfuser.sh file and change number of agents to 1 or 2 or 4 (default = 4 agents)

```
Open run_generation_transfuser.sh Save
1 #!/bin/bash
2
3 # Export other environment variables
4 export CARLA_ROOT=carla_server # path to your carla root
5 export CARLA_SERVER=${CARLA_ROOT}/CarlaUE4.sh
6 export PYTHONPATH=$PYTHONPATH:${CARLA_ROOT}/PythonAPI
7 export PYTHONPATH=$PYTHONPATH:${CARLA_ROOT}/PythonAPI/carla
8 export PYTHONPATH=$PYTHONPATH:${CARLA_ROOT}/PythonAPI/carla/dist/carla-0.9.15-py3.7-linux-x86_64
9 export PYTHONPATH=$PYTHONPATH:$(pwd -P)/leaderboard
10 export PYTHONPATH=$PYTHONPATH:$(pwd -P)/scenario_runner
11
12
13 ##### Run KING Generation #####
14 # Generate scenarios after setting the weather
15 python generate_scenarios.py \
16     --renderer_class CARLA --ego_agent transfuser \
17     --ego_agent_ckpt driving_agents/king/transfuser/model_checkpoints/regular/transfuser/ \
18     --init_root driving_agents/king/transfuser/king_initializations/initializations_subset \
19     --num_agents 4 --save_path ./generation_results_transfuser/agents_4 \
20     --opt_iters 1 --beta1 0.8 --beta2 0.99 --w_adv_col 3.0 --w_adv_rd 20.0
21
22 echo "Results"
23 echo "======"
24 python3 tools/parse_generation_results.py --results_dir ./generation_results_transfuser/agents_4 --
25     num_agents 4
```

and run:

```
bash run_generation_transfuser.sh
```

```
sanjeetha@dcs35167: ~/king
sanjeetha@dcs35167:~/king$ conda activate king
(king) sanjeetha@dcs35167:~/king$ bash run_generation_transfuser.sh
```



Collisions are also detected in the scenario using *Collision rate*

```
sanjeetha@dcs35167: ~/king
avior is equivalent to passing `weights=ResNet34_Weights.IMAGENET1K_V1`. You can
also use `weights=ResNet34_Weights.DEFAULT` to get the most up-to-date weights.
warnings.warn(msg)
0%|                               | 0/1 [00:00<?, ?it/s/
/home/sanjeetha/king/driving_agents/king/transfuser/transfuser_agent.py:171: User
Warning: The given NumPy array is not writable, and PyTorch does not support non
-writable tensors. This means writing to this tensor will result in undefined be
havior. You may want to copy the array to protect its data or make it writable b
efore converting it to a tensor. This type of warning will be suppressed for the
rest of this program. (Triggered internally at ../torch/csrc/utils/tensor_numpy
.cpp:172.)
  rgb = torch.from_numpy(scale_and_crop_image(Image.fromarray(tick_data['rgb'])),
crop=self.config.input_resolution)).unsqueeze(0)
/home/sanjeetha/anaconda3/envs/king/lib/python3.7/site-packages/numpy/matrixlib/
defmatrix.py:145: FutureWarning: The input object of type 'Tensor' is an array-l
ike implementing one of the corresponding protocols (`__array__`, `__array_inter
face__` or `__array_struct__`); but not a sequence (or 0-D). In the future, this
object will be coerced as if it was first converted using `np.array(obj)`. To r
etain the old behaviour, you have to either modify the type 'Tensor', or assign
to an empty array created with `np.empty(correct_shape, dtype=object)`.
  arr = N.array(data, dtype=dtype, copy=copy)
/home/sanjeetha/anaconda3/envs/king/lib/python3.7/site-packages/torch/functional
.py:478: UserWarning: torch.meshgrid: in an upcoming release, it will be require
d to pass the indexing argument. (Triggered internally at ../aten/src/ATen/nati
ve/TensorShape.cpp:2894.)
  return _VF.meshgrid(tensors, **kwargs) # type: ignore[attr-defined]
0%|                               | 0/1 [00:08<?, ?it/s]
Results
=====
Found 1 results files...
Collision rate: 1.0
/home/sanjeetha/anaconda3/envs/king/lib/python3.7/site-packages/numpy/core/fromn
umeric.py:3420: RuntimeWarning: Mean of empty slice.
  out=out, **kwargs)
/home/sanjeetha/anaconda3/envs/king/lib/python3.7/site-packages/numpy/core/_meth
ods.py:188: RuntimeWarning: invalid value encountered in double_scalars
  ret = ret.dtype.type(ret / rcount)
t@50: nan
-----
```

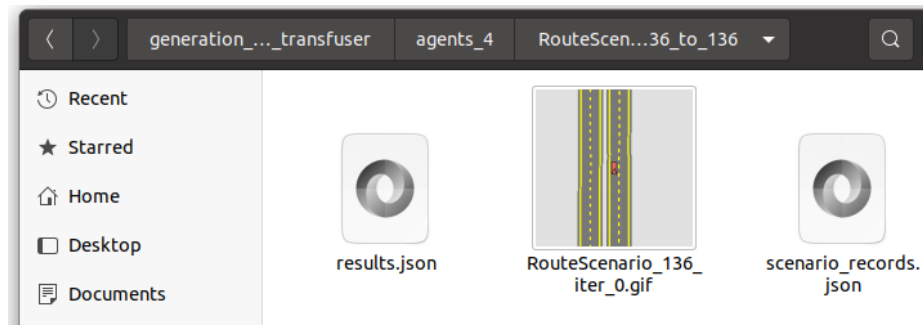
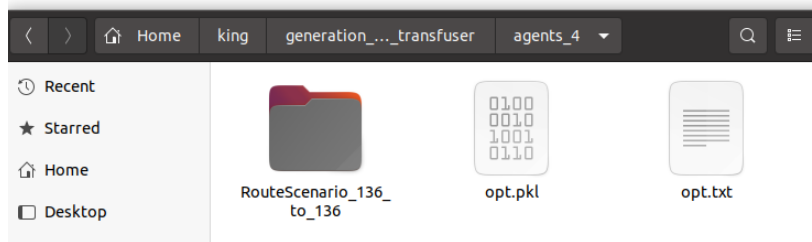
In order to visualize this scenario, run:

```
bash run_visualization.sh generation_results_transfuser
```

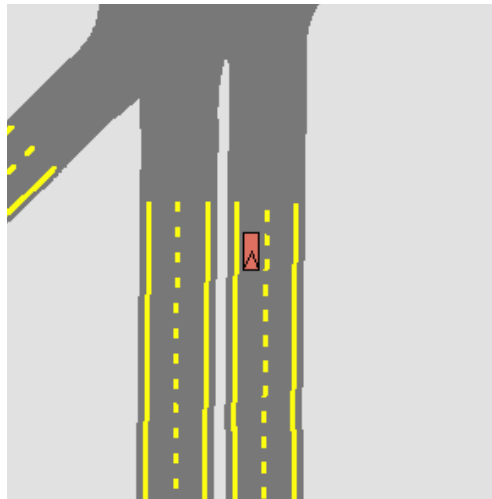
Getting results: Now you can find a new folder named *generation_results_transfuser*

```
generation_results_transfuser/
├── agents_4
│   ├── RouteScenario_136_to_136
│   │   ├── RouteScenario_136_iter_0.gif
│   │   ├── results.json
│   │   └── scenario_records.json
```

opt.pkl
opt.txt



RouteScenario_136_iter_0.gif



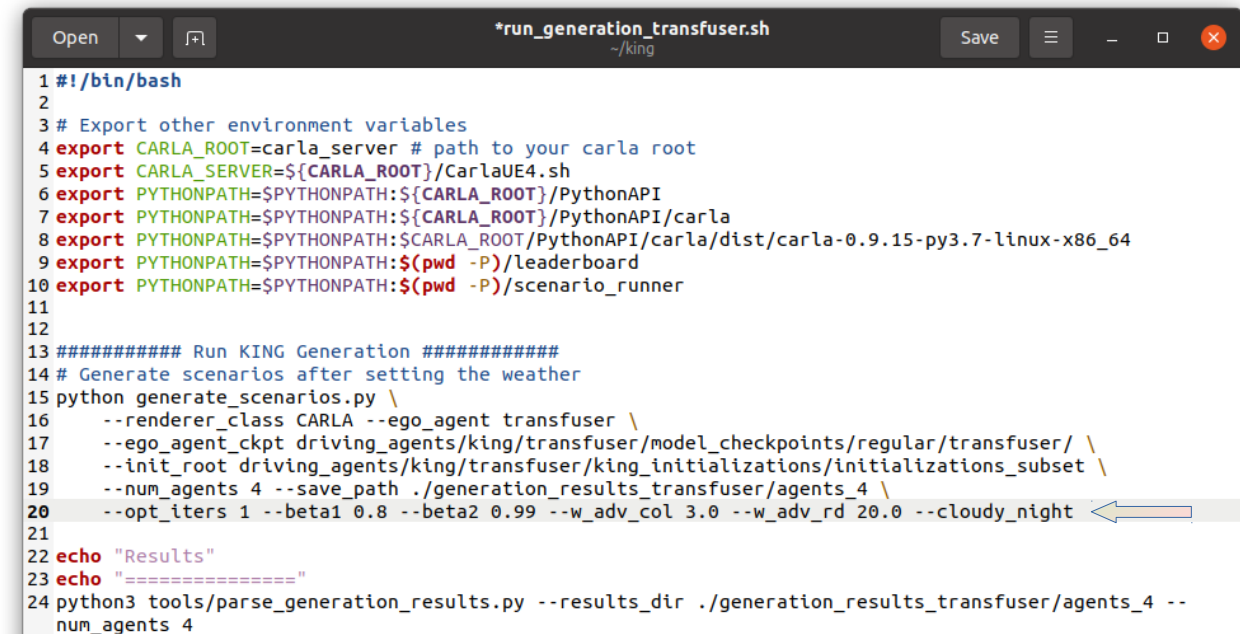
Open opt.txt to view information about scenario generated.

```
1 ===== Options =====
2 adv_col_thresh: 1.25
3 batch_size: 1
4 beta1: 0.8
5 beta2: 0.99
6 building: False
7 building_remove: False
8 cloudy_dawn: False
9 cloudy_morning: False
10 cloudy_night: False
11 cloudy_noon: False
12 cloudy_sunset: False
13 cloudy_twilight: False
14 detach_ego_path: 1
15 device: cuda
16 dynamic_weather: False
17 ego_agent: transfuser
18 ego_agentckpt: driving_agents/king/transfuser/model_checkpoints/regular/transfuser/
19 gradient_clip: 0.0
20 hard_rain_dawn: False
21 hard_rain_morning: False
22 hard_rain_night: False
23 hard_rain_noon: False
24 hard_rain_sunset: False
25 hard_rain_twilight: False
26 init_root: driving_agents/king/transfuser/king_initializations/initializations_subset
27 king_data_fps: 2
28 learning_rate: 0.005
29 max_num_routes: -1
30 mid_rain_dawn: False
31 mid_rain_morning: False
32 mid_rain_night: False
33 mid_rain_noon: False
34 mid_rain_sunset: False
35 mid_rain_twilight: False
36 num_agents: 4
37 opt_iters: 1
38 port: 2000
39 renderer_class: CARLA
40 routes_file: leaderboard/data/routes/subset_20perTown.xml
41 routes_file_adv: leaderboard/data/routes/adv_all.xml
```

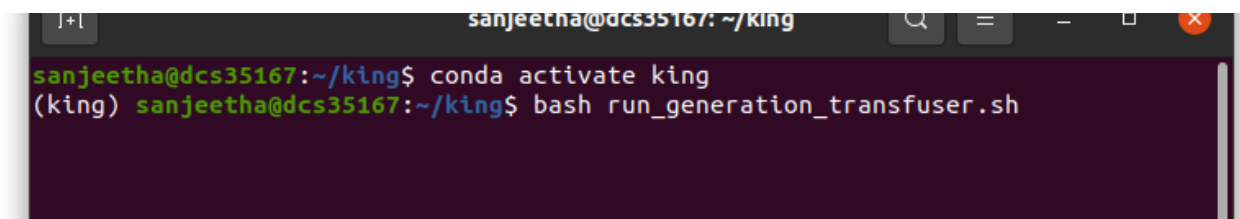
Replaying Scenario with modifications:

Open **run_generation_transfuser.sh** file, and type below arguments to tailor the environment. Add arguments as indicated in **generate_scenarios.py** file. For example if you would like to

a) change weather to **CloudyNight**: Use argument **--cloudy_night** in **run_generation_transfuser.sh**, spin up CARLA and open terminal to run **bash run_generation_transfuser.sh**

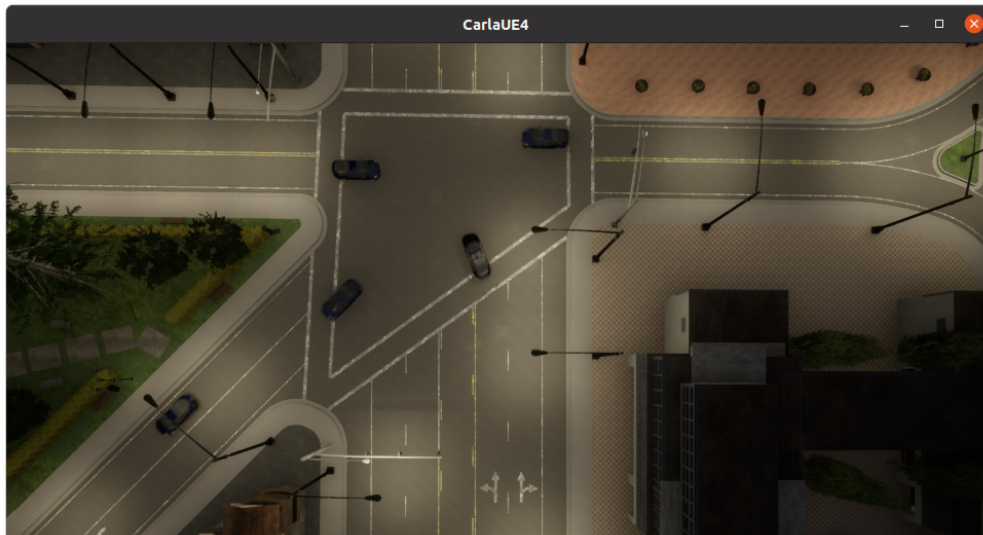


```
1 #!/bin/bash
2
3 # Export other environment variables
4 export CARLA_ROOT=carla_server # path to your carla root
5 export CARLA_SERVER=${CARLA_ROOT}/CarlaUE4.sh
6 export PYTHONPATH=$PYTHONPATH:${CARLA_ROOT}/PythonAPI
7 export PYTHONPATH=$PYTHONPATH:${CARLA_ROOT}/PythonAPI/carla
8 export PYTHONPATH=$PYTHONPATH:${CARLA_ROOT}/PythonAPI/carla/dist/carla-0.9.15-py3.7-linux-x86_64
9 export PYTHONPATH=$PYTHONPATH:$(pwd -P)/leaderboard
10 export PYTHONPATH=$PYTHONPATH:$(pwd -P)/scenario_runner
11
12
13 ##### Run KING Generation #####
14 # Generate scenarios after setting the weather
15 python generate_scenarios.py \
16     --renderer_class CARLA --ego_agent transfuser \
17     --ego_agent_ckpt driving_agents/king/transfuser/model_checkpoints/regular/transfuser/ \
18     --init_root driving_agents/king/transfuser/king_initializations/initializations_subset \
19     --num_agents 4 --save_path ./generation_results_transfuser/agents_4 \
20     --opt_iters 1 --beta1 0.8 --beta2 0.99 --w_adv_col 3.0 --w_adv_rd 20.0 --cloudy_night
21
22 echo "Results"
23 echo "====="
24 python3 tools/parse_generation_results.py --results_dir ./generation_results_transfuser/agents_4 --
25 num_agents 4
```



```
sanjeetha@dcs35167: ~/king
sanjeetha@dcs35167:~/king$ conda activate king
(king) sanjeetha@dcs35167:~/king$ bash run_generation_transfuser.sh
```

The scenario is generated with cloudy_night settings.



Similarly various arguments can be included in ***generate_scenarios.py*** to replay the selected route/scenario with modifications.

b) Toggling off all the buildings in a scenario using ***-building*** argument in ***generate_scenarios.py*** and run:

```
bash run_generation_transfuser.sh
```



c) If you want to toggle specific building in the scenario use argument ***-building_remove*** in ***generate_scenarios.py***, and select the building you want to remove by making corresponding changes to ***building_remove.py*** file



Define your location (replace these coordinates with your actual location)

location = carla.Location(x=-150.0, y=30.0, z=50.0) # to remove top left building

location = carla.Location(x=-150.0, y=70.0, z=50.0) # to remove bottom left building

location = carla.Location(x=-80.0, y=100.0, z=100.0) # to remove top right building

location = carla.Location(x=-80.0, y=180.0, z=120.0) # to remove bottom right building

Define the radius within which to search for buildings

radius = 100.0 #to remove top left building

radius = 100.0 # to remove bottom left building

radius = 150.0 # to remove top right building

radius = 200.0 # to remove bottom right building

Collect the IDs of the first n buildings

first_n_building_ids = [building.id for building in building_objects[:60]] # to remove top left building

first_n_building_ids = [building.id for building in building_objects[:100]] #to remove bottom left building

first_n_building_ids = [building.id for building in building_objects[:10]] # to remove top right building

first_n_building_ids = [building.id for building in building_objects[:45]] # to remove bottom right building

and run:

```
bash run_generation_transfuser.sh
```

